ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 C ✓ 1.1.2 A ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 F ✓ 1.3.3 H ✓ 1.3.4 I ✓ 1.3.5 E ✓ 1.3.6 A ✓ 1.3.7 J ✓ 1.3.8 B ✓ 1.3.9 G ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Setpoint input ✓ → comparator / summing point with + and − ✓ → controller ✓ → motor drive and motor ✓ → conveyor (output speed) ✓; speed sensor (tachogenerator / encoder) in the feedback path back to the − input of the comparator ✓ (6)

2.2 P – gives an output proportional to the present error, so the system responds quickly, but it leaves a small steady-state error and too much gain causes oscillation ✓✓. I – adds up the error over time and keeps changing the output until the error is zero, so it removes the steady-state error (too much causes overshoot) ✓✓. D – reacts to how fast the error is changing, so it damps the response and reduces overshoot ✓✓. (6)

2.3.1 $\%\ \text{overshoot} = (\text{peak} - \text{setpoint}) \div \text{setpoint} \times 100 = (1\,650 - 1\,500) \div 1\,500 \times 100$ ✓ $= 10\%$ ✓ (2)

2.3.2 $\text{Error} = 1\,500 - 1\,470 = 30\ \text{rpm}\ (2\%)$ ✓ (1)

2.3.3 Integral (I) action ✓ (1)

2.4 Any TWO ✓✓: a conveyor running on a timer; a washing / cleaning cycle on a timer; a fan or pump switched on at a fixed speed; a traffic-light or sequence controller on fixed times; a stepper motor driven without feedback. (2)

[18]

QUESTION 3

3.1 Strain gauges are glued to a metal beam (the load cell) ✓; the load bends the beam slightly, which stretches or compresses the gauges and changes their resistance ✓; the gauges form the arms of a Wheatstone bridge that is balanced (0 V output) with no load ✓; the load unbalances the bridge, giving a small output voltage proportional to the weight, which is amplified ✓. (4)

3.2 Sensor ✓ and reason ✓ each: (a) thermocouple (type K) – works at high temperatures, robust; (b) LVDT / linear potentiometer / magnetostrictive sensor – measures linear displacement accurately; (c) tachogenerator / encoder / inductive proximity sensor counting pulses – output proportional to speed; (d) ultrasonic / pressure (hydrostatic) / capacitive / radar level sensor – measures the level continuously without opening the tank. (8)

3.3.1 $T = (I - 4) \div 16 \times 200 = (12 - 4) \div 16 \times 200 \checkmark = 100\ °C \checkmark$ (2)

3.3.2 $I = 4 + (T \div 200) \times 16 = 4 + (150 \div 200) \times 16 \checkmark = 16\ mA \checkmark$ (2)

[16]

QUESTION 4

4.1 $\omega = 2\pi N \div 60 = 2\pi \times 1\,450 \div 60 = 151{,}84\ rad/s \checkmark$; $P = T\omega \checkmark = 25 \times 151{,}84 = 3\,796{,}09\ W\ (3{,}80\ kW) \checkmark$ (3)

4.2 $N_{out} = 1\,450 \div 20 \checkmark = 72{,}5\ rpm \checkmark$; $T_{out} = T_{in} \times ratio \times \eta = 25 \times 20 \times 0{,}9 \checkmark = 450\ N{\cdot}m \checkmark$ (4)

4.3 $A = \pi d^2 \div 4 = \pi \times 0{,}08^2 \div 4 \checkmark = 5{,}03 \times 10^{-3}\ m^2 \checkmark$; $F = pA = 10 \times 10^6 \times 5{,}03 \times 10^{-3} \checkmark = 50\,265{,}48\ N\ (50{,}27\ kN) \checkmark$ (4)

4.4 $A_{annulus} = \pi(D^2 - d^2) \div 4 = \pi(0{,}08^2 - 0{,}04^2) \div 4 \checkmark = 3{,}77 \times 10^{-3}\ m^2 \checkmark$; $F = 10 \times 10^6 \times 3{,}77 \times 10^{-3} = 37\,699{,}11\ N\ (37{,}70\ kN) \checkmark$ (3)

4.5 Hydraulic, any TWO ✓✓: very large forces from small actuators; holds a load steadily; smooth, easily controlled speed; overload protection with a relief valve. Electric, any TWO ✓✓: clean (no oil leaks); very accurate position and speed control; easy to connect to controllers; quiet and efficient; no pump or oil tank needed. (4)

4.6 $v = 0{,}5 \div 5 = 0{,}1\ m/s$; $Q = Av = 5{,}03 \times 10^{-3} \times 0{,}1 = 5{,}03 \times 10^{-4}\ m^3/s \checkmark$; $\times 60\,000 = 30{,}16\ L/min \checkmark$ (2)

[20]

QUESTION 5

5.1.1 $MTBF = \text{operating time} \div \text{number of failures} = 2\,000 \div 4 \checkmark = 500\ h \checkmark$ (2)

5.1.2 $MTTR = \text{total repair time} \div \text{number of repairs} = 40 \div 4 \checkmark = 10\ h \checkmark$ (2)

5.1.3 $\text{Availability} = \text{MTBF} \div (\text{MTBF} + \text{MTTR}) \times 100 = 500 \div 510 \times 100$ ✓ = $98{,}04\%$ ✓ (2)

5.2 Preventive maintenance is done at fixed intervals (time or operating hours) whether or not the part shows wear ✓, e.g. changing the gearbox oil every 2 000 hours ✓. Predictive maintenance is done when measurements show that a part is starting to fail ✓, e.g. replacing a bearing when vibration analysis or thermal imaging shows a fault developing ✓. (4)

5.3 $R = R_1 \times R_2 \times R_3$ ✓ $= 0{,}95^3$ ✓ $= 0{,}86$ $(85{,}74\%)$ ✓ (3)

5.4 $R = 1 - (1 - 0{,}9)^2 = 1 - 0{,}01$ ✓ $= 0{,}99$ (99%) ✓; the system fails only if BOTH pumps fail at the same time, which is much less likely than one pump failing (redundancy) ✓ (3)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE